

National Load Despatch Centre
राष्ट्रीय भार प्रेषण केंद्र
GRID CONTROLLER OF INDIA LIMITED
ग्रिड कंट्रोलर ऑफ इंडिया लिमिटेड

(Government of India Enterprise/ भारत सरकार का उद्यम)
B-9, QUTUB INSTITUTIONAL AREA, KATWARIA SARAI, NEW DELHI -110016
बी-9, कुतुब इंस्टीट्यूशनल एरिया, कटवारिया सराये, न्यू दिल्ली-110016

Ref: POSOCO/NLDC/SO/Daily PSP Report

दिनांक: 19th April 2024

To,

- कार्यकारी निदेशक, पू.क्षे.भा.प्रे.के., 14, गोल्फ क्लब रोड, कोलकाता - 700033
Executive Director, ERLDC, 14 Golf Club Road, Tollygunge, Kolkata, 700033
- कार्यकारी निदेशक, ऊ.क्षे.भा.प्रे.के., 18/ए, शाहीद जीत सिंह सनसनवाल मार्ग, नई दिल्ली - 110016
Executive Director, NRLDC, 18-A, Shaheed Jeet Singh Marg, Katwaria Sarai, New Delhi - 110016
- कार्यकारी निदेशक, प.क्षे.भा.प्रे.के., एफ3, एम आई डी सी क्षेत्र, अंधेरी, मुंबई -400093
Executive Director, WRLDC, F-3, M.I.D.C. Area, Marol, Andheri (East), Mumbai-400093
- कार्यकारी निदेशक, ऊ.पू.क्षे.भा.प्रे.के., डोंगतेह, लोअर नोंग्रह, लापालंग, शिलोंग - 793006
Executive Director, NERLDC, Dongteih, Lower Nongrah, Lapalang, Shillong - 793006, Meghalaya
- कार्यकारी निदेशक, द.क्षे.भा.प्रे.के., 29, रेस कोर्स क्रॉस रोड, बंगलुरु -560009
Executive Director, SRLDC, 29, Race Course Cross Road, Bangalore-560009

Sub: Daily PSP Report for the date 18.04.2024.

महोदय/Dear Sir,

आईईजीसी-2010 की धारा स.-5.5.1 के प्रावधान के अनुसार, दिनांक 18-अप्रैल-2024 की अखिल भारतीय प्रणाली की दैनिक ग्रिड निष्पादन रिपोर्ट रांभांप्रेके की वेबसाइट पर उपलब्ध है।

As per article 5.5.1 of the Indian Electricity Grid Code, the daily report pertaining power supply position of All India Power System for the date 18th April 2024, is available at the NLDC website.

धन्यवाद,

Report for previous day Date of Reporting: 19-Apr-2024

A. Power Supply Position at All India and Regional level

	NR	WR	SR	ER	NER	TOTAL
Demand Met during Evening Peak hrs(MW) (at 20:00 hrs; from RLDCs)	58237	66509	52748	27713	2778	207985
Peak Shortage (MW)	0	0	0	0	0	0
Energy Met (MU)	1241	1615	1368	631	51	4907
Hydro Gen (MU)	167	42	52	31	14	305
Wind Gen (MU)	34	98	46	-	-	178
Solar Gen (MU)*	117.68	91.53	125.16	3.14	0.91	338
Energy Shortage (MU)	0.00	0.00	0.00	0.98	0.00	0.98
Maximum Demand Met During the Day (MW) (From NLDC SCADA)	58831	72684	65684	29074	2964	214072
Time Of Maximum Demand Met	19:45	15:44	14:59	23:33	18:23	22:38

B. Frequency Profile (%)

Region	FVI	< 49.7	49.7 - 49.8	49.8 - 49.9	< 49.9	49.9 - 50.05	> 50.05
All India	0.051	0.24	0.27	5.30	5.81	71.61	22.58

C. Power Supply Position in States

Region	States	Max.Demand Met during the day (MW)	Shortage during maximum Demand (MW)	Energy Met (MU)	Drawal Schedule (MU)	OD(+)/UD(-) (MU)	Max OD (MW)	Energy Shortage (MU)
NR	Punjab	8605	0	158.5	47.5	0.5	268	0.00
	Haryana	8501	0	160.7	97.4	-0.3	220	0.00
	Rajasthan	13105	0	268.1	45.1	-4.1	455	0.00
	Delhi	4780	0	97.2	87.4	1.0	262	0.00
	UP	22301	0	420.5	174.5	-1.4	544	0.00
	Uttarakhand	2189	0	43.9	24.2	-2.1	75	0.00
	HP	1662	0	32.1	15.6	0.0	51	0.00
	J&K(UT) & Ladakh(UT)	2442	0	51.0	37.3	0.1	253	0.00
	Chandigarh	235	0	4.7	4.8	0.0	14	0.00
WR	Railways_NR ISTS	206	0	4.2	3.7	0.5	70	0.00
	Chhattisgarh	6116	0	138.9	77.8	-0.4	281	0.00
	Gujarat	23289	0	489.2	165.2	-0.3	721	0.00
	MP	12506	0	273.8	140.0	-4.6	403	0.00
	Maharashtra	28705	0	635.0	216.1	-2.1	790	0.00
	Goa	801	0	17.2	12.2	4.7	47	0.00
	DNHDDPDCL	1281	0	29.4	29.5	-0.1	27	0.00
	AMNSIL	846	0	18.9	5.1	0.6	291	0.00
	BALCO	520	0	12.4	12.5	-0.1	7	0.00
SR	Andhra Pradesh	13229	0	252.6	95.8	1.5	1227	0.00
	Telangana	11784	0	243.0	117.6	-0.1	521	0.00
	Karnataka	16541	0	316.7	153.2	0.2	789	0.00
	Kerala	5355	0	108.4	84.1	1.3	426	0.00
	Tamil Nadu	20341	0	436.7	264.7	-2.7	498	0.00
	Puducherry	484	0	11.0	10.6	-0.3	33	0.00
ER	Bihar	6559	0	129.5	120.4	-1.9	378	0.74
	DVC	3310	0	74.0	-38.8	0.9	341	0.00
	Jharkhand	1968	0	39.3	32.2	-3.5	389	0.24
	Odisha	6463	0	139.7	56.9	-1.0	466	0.00
	West Bengal	11737	0	246.9	98.8	-2.1	692	0.00
	Sikkim	87	0	1.6	1.6	0.0	27	0.00
NER	Railways_ER ISTS	10	0	0.1	0.1	0.0	0	0.00
	Arunachal Pradesh	135	0	2.0	2.7	-0.8	37	0.00
	Assam	1842	0	33.6	27.1	-0.1	148	0.00
	Manipur	208	0	2.9	3.1	-0.2	22	0.00
	Meghalaya	335	0	5.5	4.2	-0.3	245	0.00
	Mizoram	101	0	1.5	1.7	-0.4	14	0.00
	Nagaland	151	0	2.3	2.3	-0.1	15	0.00
	Tripura	257	0	3.5	5.0	0.0	125	0.00

D. Transnational Exchanges (MU) - Import(+ve)/Export(-ve)

	Bhutan	Nepal	Bangladesh	Godda -> Bangladesh
Actual (MU)	4.5	-9.5	-24.3	-25.0
Day Peak (MW)	427.0	-673.1	-1072.0	-1330.0

E. Import/Export by Regions (in MU) - Import(+ve)/Export(-ve); OD(+)/UD(-)

	NR	WR	SR	ER	NER	TOTAL
Schedule(MU)	108.8	-347.9	263.0	-31.3	7.4	0.0
Actual(MU)	109.5	-367.0	274.6	-39.9	8.4	-14.4
O/D/U/D(MU)	0.8	-19.2	11.6	-8.6	1.0	-14.4

F. Generation Outage(MW)

	NR	WR	SR	ER	NER	TOTAL	% Share
Central Sector	2668	4429	3685	3616	155	14553	46
State Sector	6901	4841	3068	2384	196	17389	54
Total	9569	9270	6753	5999	351	31941	100

G. Sourcewise generation (Gross) (MU)

	NR	WR	SR	ER	NER	All India	% Share
Coal	804	1713	777	727	15	4035	76
Lignite	23	19	73	0	0	115	2
Hydro	167	42	52	31	14	305	6
Nuclear	25	64	68	0	0	158	3
Gas, Naptha & Diesel	36	87	6	0	20	149	3
RES (Wind, Solar, Biomass & Others)	163	193	203	5	1	565	11
Total	1219	2117	1179	762	50	5327	100

Share of RES in total generation (%)	13.41	9.11	17.23	0.62	1.86	10.61
Share of Non-fossil fuel (Hydro,Nuclear and RES) in total generation(%)	29.17	14.12	27.40	4.71	29.25	19.30

H. All India Demand Diversity Factor

Based on Regional Max Demands	1.070
Based on State Max Demands	1.116

I. All India Peak Demand and shortage at Solar and Non-Solar Hour

	Max Demand Met(MW)	Time	Shortage(MW)
Solar hr	213454	15:52	0
Non-Solar hr	214072	22:38	0

Diversity factor = Sum of regional or state maximum demands / All India maximum demand

**Note: All generation MU figures are gross

***Godda (Jharkhand) -> Bangladesh power exchange is through the radial connection (isolated from Indian Grid)

Solar Hours -> 07:00 to 17:00hrs and rest are Non-Solar Hours

*Source: RLDCs for solar connected to ISTS; SLDCs for embedded solar. Limited visibility of embedded solar data.

Executive Director-NLDC

INTER-REGIONAL EXCHANGES

Import=(+ve) /Export =(-ve) for NET (MU)

Date of Reporting: 19-Apr-2024

Sl No	Voltage Level	Line Details	No. of Circuit	Max Import (MW)	Max Export (MW)	Import (MU)	Export (MU)	NET (MU)
Import/Export of ER (With NR)								
1	HVDC	ALIPURDUAR-AGRA	2	0	0	0.0	0.0	0.0
2	HVDC	PUSAULI B/B	-	0	48	0.0	1.3	-1.3
3	765 kV	GAYA-VARANASI	2	800	255	2.4	0.0	2.4
4	765 kV	SASARAM-FATEHPUR	1	107	271	0.0	3.5	-3.5
5	765 kV	GAYA-BALIA	1	204	472	0.0	5.1	-5.1
6	400 kV	PUSAULI-VARANASI	1	0	104	0.0	1.5	-1.5
7	400 kV	PUSAULI -ALLAHABAD	1	58	14	0.3	0.0	0.3
8	400 kV	MUZAFFARPUR-GORAKHPUR	2	804	224	3.9	0.0	3.9
9	400 kV	PATNA-BALIA	2	435	643	0.0	7.1	-7.1
10	400 kV	NAUBATPUR-BALIA	2	11	192	0.0	1.8	-1.8
11	400 kV	BIHARSHARIFF-BALIA	2	365	29	4.2	0.0	4.2
12	400 kV	MOTIHARI-GORAKHPUR	2	346	312	0.0	1.7	-1.7
13	400 kV	BIHARSHARIFF-VARANASI	2	228	43	1.1	0.0	1.1
14	220 kV	SAHUPURI-KARAMNASA	1	57	74	0.0	0.6	-0.6
15	132 kV	NAGAR UNTARI-RIHAND	1	0	0	0.0	0.0	0.0
16	132 kV	GARWAH-RIHAND	1	30	0	0.7	0.0	0.7
17	132 kV	KARMANASA-SAHUPURI	1	0	0	0.0	0.0	0.0
18	132 kV	KARMANASA-CHANDAULI	1	0	0	0.0	0.0	0.0
ER-NR						12.6	22.7	-10.1
Import/Export of ER (With WR)								
1	765 kV	JHARSUGUDA-DHARAMJAIGARH	4	1754	0	25.8	0.0	25.8
2	765 kV	NEW RANCHI-DHARAMJAIGARH	2	1145	92	13.5	0.0	13.5
3	765 kV	JHARSUGUDA-DURG	2	0	802	0.0	12.4	-12.4
4	400 kV	JHARSUGUDA-RAIGARH	4	0	347	0.0	4.5	-4.5
5	400 kV	RANCHI-SIPAT	2	215	84	1.2	0.0	1.2
6	220 kV	BUDHIPADAR-RAIGARH	1	61	91	0.0	0.5	-0.5
7	220 kV	BUDHIPADAR-KORBA	2	0	2	0.0	0.0	0.0
ER-WR						40.4	17.3	23.1
Import/Export of ER (With SR)								
1	HVDC	JEYPORE-GAZUWAKA B/B	2	0	679	0.0	14.2	-14.2
2	HVDC	TALCHER-KOLAR BIPOLE	2	0	1486	0.0	36.1	-36.1
3	765 kV	ANGUL-SRIKAKULAM	2	0	2812	0.0	49.8	-49.8
4	400 kV	TALCHER-I/C	2	428	0	8.8	0.0	8.8
5	220 kV	BALIMELA-UPPER-SILERRU	1	0	0	0.0	0.0	0.0
ER-SR						0.0	100.1	-100.1
Import/Export of ER (With NER)								
1	400 kV	BINAGURI-BONGAIGAON	2	292	308	2.1	1.9	0.1
2	400 kV	ALIPURDUAR-BONGAIGAON	2	480	399	1.6	0.0	1.6
3	220 kV	ALIPURDUAR-SALAKATI	2	55	48	0.1	0.0	0.1
ER-NER						3.8	1.9	1.8
Import/Export of NER (With NR)								
1	HVDC	BISWANATH CHARIALI-AGRA	2	364	0	9.0	0.0	9.0
NER-NR						9.0	0.0	9.0
Import/Export of WR (With NR)								
1	HVDC	CHAMPA-KURUKSHETRA	2	0	2011	0.0	47.8	-47.8
2	HVDC	VINDHYACHAL B/B	-	493	0	12.1	0.0	12.1
3	HVDC	MUNDRA-MOHINDERGARH	2	0	1172	0.0	18.8	-18.8
4	765 kV	GWALIOR-AGRA	2	226	1882	0.1	21.9	-21.8
5	765 kV	GWALIOR-PHAGI	2	1105	909	3.9	7.2	-3.4
6	765 kV	JABALPUR-ORAI	2	412	690	0.0	11.2	-11.2
7	765 kV	GWALIOR-ORAI	1	547	0	7.8	0.0	7.8
8	765 kV	SATNA-ORAI	1	0	992	0.0	18.9	-18.9
9	765 kV	BANASKANTHA-CHITORGARH	2	1375	449	7.6	1.6	6.0
10	765 kV	VINDHYACHAL-VARANASI	2	0	2833	0.0	41.8	-41.8
11	400 kV	ZERDA-KANKROLI	1	323	3	2.8	0.0	2.8
12	400 kV	ZERDA -BHINMAL	1	322	2	2.8	0.0	2.8
13	400 kV	VINDHYACHAL -RIHAND	1	931	0	12.0	0.0	12.0
14	400 kV	RAPP-SHUJALPUR	2	712	181	4.7	0.5	4.2
15	400 kV	NEEMUCH-Chittorgarh	2	649	131	4.5	0.3	4.2
16	220 kV	BHANPURA-RANPUR	1	0	126	0.0	2.3	-2.3
17	220 kV	BHANPURA-MORAK	1	0	30	0.0	2.0	-2.0
18	220 kV	MEHGAON-AURAIYA	1	94	7	1.3	0.0	1.3
19	220 kV	MALANPUR-AURAIYA	1	85	0	1.0	0.0	1.0
20	132 kV	GWALIOR-SAWAI MADHOPUR	1	0	0	0.0	0.0	0.0
21	132 kV	RAJGHAT-LALITPUR	2	0	0	0.0	0.0	0.0
WR-NR						60.6	174.3	-113.7
Import/Export of WR (With SR)								
1	HVDC	BHADRAWATI B/B	-	0	1011	0.0	21.0	-21.0
2	HVDC	RAIGARH-PUGALUR	2	0	6629	0.0	97.4	-97.4
3	765 kV	SOLAPUR-RAICHUR	2	0	1874	0.0	26.6	-26.6
4	765 kV	WARDHA-NIZAMABAD	2	0	2632	0.0	47.7	-47.7
5	765 kV	WARORA-WARANGAL(NEW)	2	0	2857	0.0	53.7	-53.7
6	400 kV	KOLHAPUR-KUDGI	2	1158	0	14.9	0.0	14.9
7	220 kV	KOLHAPUR-CHIKODI	2	0	0	0.0	0.0	0.0
8	220 kV	PONDA-AMBEWADI	1	0	88	0.0	1.7	-1.7
9	220 kV	XELDEM-AMBEWADI	1	0	141	2.7	0.0	2.7
WR-SR						17.6	248.1	-230.4

INTERNATIONAL EXCHANGES					Import(+ve)/Export(-ve)	
State	Region	Line Name	Max (MW)	Min (MW)	Avg (MW)	Energy Exchange (MU)
BHUTAN	ER	400kV MANGDECHHU-ALIPURDUAR 1,2&3 i.e. ALIPURDUAR RECEIPT (from MANGDECHU HEP 4*180MW)	235	-65	87	2.08
	ER	400kV TALA-BINAGURI 1,2,4 (& 400kV MALBASE - BINAGURI) i.e. BINAGURI RECEIPT (from TALA HEP 6*170MW)	268	34	152	3.65
	ER	220kV CHUKHA-BIRPARA 1&2 (& 220kV MALBASE - BIRPARA) i.e. BIRPARA RECEIPT (from CHUKHA HEP 4*84MW)	-133	-59	-91	-2.18
	NER	132kV GELEPHU-SALAKATI	15	-8	4	0.09
	NER	132kV MOTANGA-RANGIA	41	25	36	0.86
NEPAL	NR	NEPAL IMPORT (FROM UP)	-70	0	-45	-1.08
	NR	132kV MAHENDRANAGAR-TANAKPUR(NHPC)	-70	0	-37	-0.88
	ER	NEPAL IMPORT (FROM BIHAR)	-190	0	-76	-1.83
	ER	400kV DHALKEBAR-MUZAFFARPUR 1&2	-483	-1	-240	-5.75
BANGLADESH	ER	BHERAMARA B/B HVDC (B'DESH)	-956	-793	-910	-21.83
	ER (Isolated from Indian Grid)	400kV GODDA_TPS-RAHANPUR (B'DESH) D/C	-1330	-807	-1042	-25.00
	NER	132kV COMILLA-SURAJMANI NAGAR 1&2	-116	0	-103	-2.48

Export From India (in MU)

Country	GNA (ISGS/PPA)	T-GNA							TOTAL
		BILATERAL TOTAL	COLLECTIVE						
			IDAM			RTM			
			IEX	PXIL	HPX	IEX	PXIL	HPX	
Bhutan	0.00	0.00	1.78	0.00	0.00	0.00	0.00	0.00	1.78
Nepal	0.24	0.00	6.37	0.00	0.00	0.00	0.00	0.00	6.61
Bangladesh	21.76	0.00	0.00	0.00	0.00	0.00	0.00	0.00	21.76
Myanmar	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Total Export	22.00	0.00	8.15	0.00	0.00	0.00	0.00	0.00	30.15

Import by India(in MU)

Country	GNA (ISGA/PPA)	T-GNA							TOTAL
		BILATERAL TOTAL	COLLECTIVE						
			IDAM			RTM			
			IEX	PXIL	HPX	IEX	PXIL	HPX	
Bhutan	5.18	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.18
Nepal	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Bangladesh	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Myanmar	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Total Import	5.18	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.18

Net from India(in MU)

-ve : Export / +ve : Import

Country	GNA (ISGS/PPA)	T-GNA							TOTAL
		BILATERAL TOTAL	COLLECTIVE						
			IDAM			RTM			
			IEX	PXIL	HPX	IEX	PXIL	HPX	
Bhutan	5.18	0.00	-1.78	0.00	0.00	0.00	0.00	0.00	3.40
Nepal	-0.24	0.00	-6.37	0.00	0.00	0.00	0.00	0.00	-6.61
Bangladesh	-21.76	0.00	0.00	0.00	0.00	0.00	0.00	0.00	-21.76
Myanmar	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Total Net	-16.82	0.00	-8.15	0.00	0.00	0.00	0.00	0.00	-24.97